

```
1 /Users/mehakagrawal/PycharmProjects/  
  Final_Dissertation/.venv/bin/python /Users/  
  mehakagrawal/Desktop/Final_Dissertation/Codes/7.  
  string_prep_and_grn_overlap/  
  string_prep_and_grn_overlap.py  
2 STRING input preparation + GRN cross-reference  
3 Significant DEGs (q<0.05, |log2FC|>1.0): 5541 (  
  4252 down, 1289 up in MUT)  
4   all : 5541 DEGs -> 3594 mapped -> 3698 human  
  symbols (string_input_human_symbols.txt)  
5   up  : 1289 DEGs -> 859 mapped -> 917 human  
  symbols (string_input_UP_in_MUT_human.txt)  
6   down: 4252 DEGs -> 2735 mapped -> 2879 human  
  symbols (string_input_DOWN_in_MUT_human.txt)  
7  
8 GRN cross-reference  
9 DOWN  'Cell Cycle' (Reactome, 248 genes): 65  
  repressed-TF targets, 65 E2F targets  
10 UP    'Response to stimulus' (GO Process, 497 genes  
  ): 252 activated-TF targets, 62 TP53 targets  
11   Saved: grn_string_overlap_summary.csv  
12  
13 Process finished successfully! Outputs in: /Users/  
  mehakagrawal/Desktop/Final_Dissertation/Outputs/7.  
  string_prep_and_grn_overlap/  
14  
15 Process finished with exit code 0  
16
```